

Lecture 20: Self-gravitation and rotation

David Al-Attar

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Outline and motivation

In previous lectures we have discussed how ~~that~~ travel time and waveform observations can be used to determine variations in elastic wave speeds within the Earth. Nothing has, however, been said about **density**, with this parameter being key to understanding both the Earth's interior composition and dynamics. To learn about density, we need to study the ~~long-period free oscillations~~ long-period free oscillations of the Earth. This is because **self-gravitation**, and hence density, plays a vital role in their physics. Within this lecture we will return to basic elastodynamics, and show how gravitational forces can be included into the equations of motion. We also explain how the **Earth's rotation** enters into the problem. Having obtained the equations of motion, we perform a **scaling analysis** to determine the relative importance of the different terms, and conclude that at short enough periods the effects of self-gravitation and rotation can be safely neglected, and hence the earlier parts of this course remain valid.

A review of Newtonian gravitation

Consider the motion φ of an elastic body relative to a reference body M , and write M_t for the instantaneous volume of space occupied at time t . At this instant in time, the gravitational potential of the body satisfies Poisson~~equation~~

$$\nabla^2 \phi = 4\pi G \varrho.$$

's equation

$$\nabla^2 \phi = 4\pi G \varrho, \tag{1}$$

in $\mathbb{R}^3$, where G is the universal gravitational constant, and ϱ is the density. The gravitational acceleration associated with this potential is given by

$$g_i = - \frac{\partial \phi}{\partial x_i}$$

$$\nabla^2 \phi = 4\pi G \varrho. \tag{2}$$

The above Poisson's equation is subject to the conditions that ϕ tends to zero at infinity, while across any jumps in density we have the continuity conditions

$$\begin{aligned} [\phi]_{-}^{+} &= 0, \quad [g_i \hat{n}_i]_{-}^{+} = 0 \\ [\phi]_{-}^{+} &= 0, \quad [g_i \hat{n}_i]_{-}^{+} = 0, \end{aligned} \quad (3)$$

where $[\cdot]_{-}^{+}$ denotes the jump in a quantity in the direction of the outward normal. The closed form solution to this problem is given by

$$\begin{aligned} \phi(\mathbf{y}, t) &= -G \int_{M_t} \frac{\varrho(\mathbf{y}', t)}{[(y_i - y'_i)(y_i - y'_i)]^{\frac{1}{2}}} d^3 \mathbf{y}', \\ \phi(\mathbf{y}, t) &= -G \int_{M_t} \frac{\varrho(\mathbf{y}', t)}{[(y_i - y'_i)(y_i - y'_i)]^{\frac{1}{2}}} d^3 \mathbf{y}', \end{aligned} \quad (4)$$

and differentiating, we similar find

$$g_i(\mathbf{y}, t) = -G \int_{M_t} \frac{\varrho(\mathbf{y}', t)(y_i - y'_i)}{[(y_j - y'_j)(y_j - y'_j)]^{\frac{3}{2}}} d^3 \mathbf{y}'.$$

similarly find

$$g_i(\mathbf{y}, t) = -G \int_{M_t} \frac{\varrho(\mathbf{y}', t)(y_i - y'_i)}{[(y_j - y'_j)(y_j - y'_j)]^{\frac{3}{2}}} d^3 \mathbf{y}'. \quad (5)$$

It will be convenient to define the **referential gravitational potential** by

$$\begin{aligned} \zeta(\mathbf{x}, t) &= \phi[\varphi(\mathbf{x}, t), t], \\ \zeta(\mathbf{x}, t) &= \phi[\varphi(\mathbf{x}, t), t], \end{aligned} \quad (6)$$

which is a time-dependent field on the reference body, M . We recall from Lecture 12 13 that the volume elements in the reference body and physical space are related by $d^3 \mathbf{y} = J(\mathbf{x}, t) d^3 \mathbf{x}$ with $J = \det(\mathbf{F})$ the Jacobian of the motion. We can then change variables within eq. (4) and use the definition of ζ in eq. (6) to obtain

$$\begin{aligned} \zeta(\mathbf{x}, t) &= -G \int_M \frac{\rho(\mathbf{x}')}{\{[\varphi_i(\mathbf{x}, t) - \varphi_i(\mathbf{x}', t)][\varphi_i(\mathbf{x}, t) - \varphi_i(\mathbf{x}', t)]\}^{\frac{1}{2}}} d^3 \mathbf{x}', \\ \zeta(\mathbf{x}, t) &= -G \int_M \frac{\rho(\mathbf{x}')}{\{[\varphi_i(\mathbf{x}, t) - \varphi_i(\mathbf{x}', t)][\varphi_i(\mathbf{x}, t) - \varphi_i(\mathbf{x}', t)]\}^{\frac{1}{2}}} d^3 \mathbf{x}', \end{aligned} \quad (7)$$

where from Lecture 12 13 we know that conservation of mass requires

$$\rho(\mathbf{x}) = J(\mathbf{x}, t) \varrho[\varphi(\mathbf{x}, t), t],$$

$$\rho(\mathbf{x}) = J(\mathbf{x}, t) \varrho[\varphi(\mathbf{x}, t), t], \quad (8)$$

with ρ the **referential density**. In a similar manner, we define the **referential gravitational acceleration** by

$$\begin{aligned} \gamma_i(\mathbf{x}, t) &= g_i[\varphi(\mathbf{x}, t), t] \\ \gamma_i(\mathbf{x}, t) &= g_i[\varphi(\mathbf{x}, t), t], \end{aligned} \quad (9)$$

which from eq.(5) can be written

$$\begin{aligned} \gamma_i(\mathbf{x}, t) &= -G \int_M \frac{\rho(\mathbf{x}') [\varphi_i(\mathbf{x}, t) - \varphi_i(\mathbf{x}', t)]}{\{[\varphi_j(\mathbf{x}, t) - \varphi_j(\mathbf{x}', t)][\varphi_j(\mathbf{x}, t) - \varphi_j(\mathbf{x}', t)]\}^{\frac{3}{2}}} d\mathbf{x}', \\ \gamma_i(\mathbf{x}, t) &= -G \int_M \frac{\rho(\mathbf{x}') [\varphi_i(\mathbf{x}, t) - \varphi_i(\mathbf{x}', t)]}{\{[\varphi_j(\mathbf{x}, t) - \varphi_j(\mathbf{x}', t)][\varphi_j(\mathbf{x}, t) - \varphi_j(\mathbf{x}', t)]\}^{\frac{3}{2}}} d^3\mathbf{x}'. \end{aligned} \quad (10)$$

Gravitational binding energy

At each instant of time we can associate a **gravitational binding energy** with the elastic body. This is the energy that would be required to assemble the body from matter dispersed at infinity, assuming that its constituent parts are acted on only by their mutual gravitational attraction. Because gravity is a conservative force, the binding energy is independent of the details of the assembly.

Suppose that the mass of the body is already in place, and that its spatial density and gravitational potential are denoted by ϱ and ϕ . To determine an expression for the binding energy we will consider how this quantity changes due to an infinitesimal addition of mass to the body. By definition of the gravitational potential, the work required to bring in an additional mass element $\delta\varrho d^3\mathbf{y}$ from infinity to a point $\mathbf{y}$ within the body is $\delta\varrho \phi d^3\mathbf{y}$. The **total change in the** binding energy associated with a perturbation $\delta\varrho$ to **the** body's density **by** is, therefore, given by

$$\begin{aligned} \delta\mathcal{V}_g &= \int_{M_t} \delta\varrho \phi d^3\mathbf{y}. \\ \delta\mathcal{V}_g &= \int_{M_t} \delta\varrho \phi d^3\mathbf{y}, \end{aligned} \quad (11)$$

to first-order accuracy. Using the perturbed Poisson's equation $\nabla^2 \delta\phi = 4\pi G \delta\varrho$ for the resulting change in gravitational potential, we can write this increment to the binding energy as

$$\begin{aligned} \delta\mathcal{V}_g &= \frac{1}{4\pi G} \int_{M_t} \left(\frac{\partial^2 \delta\phi}{\partial x_i \partial x_i} \right) \phi d^3\mathbf{y} = \frac{1}{4\pi G} \int_{\mathbb{R}^3} \left(\frac{\partial^2 \delta\phi}{\partial x_i \partial x_i} \right) \phi d^3\mathbf{y}, \\ \delta\mathcal{V}_g &= \frac{1}{4\pi G} \int_{M_t} \left(\frac{\partial^2 \delta\phi}{\partial x_i \partial x_i} \right) \phi d^3\mathbf{y} = \frac{1}{4\pi G} \int_{\mathbb{R}^3} \left(\frac{\partial^2 \delta\phi}{\partial x_i \partial x_i} \right) \phi d^3\mathbf{y}, \end{aligned} \quad (12)$$

where the final equality follows from the fact that $\nabla^2 \delta\phi = 0$ outside of the instantaneous body M_t . Using the identity

$$\begin{aligned} \left(\frac{\partial^2 \delta\phi}{\partial x_i \partial x_i} \right) \phi &= \frac{\partial}{\partial x_i} \left(\phi \frac{\partial \delta\phi}{\partial x_i} \right) - \frac{\partial \phi}{\partial x_i} \frac{\partial \delta\phi}{\partial x_i} \\ \left(\frac{\partial^2 \delta\phi}{\partial x_i \partial x_i} \right) \phi &= \frac{\partial}{\partial x_i} \left(\phi \frac{\partial \delta\phi}{\partial x_i} \right) - \frac{\partial \phi}{\partial x_i} \frac{\partial \delta\phi}{\partial x_i}, \end{aligned} \quad (13)$$

and applying the divergence theorem we obtain

$$\begin{aligned} \delta\mathcal{V}_g &= -\frac{1}{4\pi G} \int_{\mathbb{R}^3} \frac{\partial \phi}{\partial x_i} \frac{\partial \delta\phi}{\partial x_i} d^3\mathbf{y}. \\ \delta\mathcal{V}_g &= -\frac{1}{4\pi G} \int_{\mathbb{R}^3} \frac{\partial \phi}{\partial x_i} \frac{\partial \delta\phi}{\partial x_i} d^3\mathbf{y}, \end{aligned} \quad (14)$$

where we have used the continuity of the potentials and their normal derivatives across ∂M_t , while the surface integral at infinity vanishes because $\phi \sim r^{-1}$ and $\|\nabla \delta\phi\| \sim r^{-2}$ as $r \rightarrow \infty$. It follows that the increment to the gravitational binding energy can be written

$$\begin{aligned} \delta\mathcal{V}_g &= \delta \left(-\frac{1}{8\pi G} \int_{\mathbb{R}^3} \frac{\partial \phi}{\partial x_i} \frac{\partial \phi}{\partial x_i} d^3\mathbf{y} \right) \\ \delta\mathcal{V}_g &= \delta \left(-\frac{1}{8\pi G} \int_{\mathbb{R}^3} \frac{\partial \phi}{\partial x_i} \frac{\partial \phi}{\partial x_i} d^3\mathbf{y} \right), \end{aligned} \quad (15)$$

and so we arrive at the result

$$\begin{aligned} \mathcal{V}_g &= -\frac{1}{8\pi G} \int_{\mathbb{R}^3} \frac{\partial \phi}{\partial x_i} \frac{\partial \phi}{\partial x_i} d^3\mathbf{y}. \\ \mathcal{V}_g &= -\frac{1}{8\pi G} \int_{\mathbb{R}^3} \frac{\partial \phi}{\partial x_i} \frac{\partial \phi}{\partial x_i} d^3\mathbf{y}. \end{aligned} \quad (16)$$

Using the Poisson's equation for ϕ , we can write the integrand of the above expression in the form

$$\begin{aligned} \frac{\partial \phi}{\partial x_i} \frac{\partial \phi}{\partial x_i} &= \frac{\partial}{\partial x_i} \left(\phi \frac{\partial \phi}{\partial x_i} \right) - \phi \left(\frac{\partial^2 \phi}{\partial x_i \partial x_i} \right) = \frac{\partial}{\partial x_i} \left(\phi \frac{\partial \phi}{\partial x_i} \right) - 4\pi G \rho \phi, \\ \frac{\partial \phi}{\partial x_i} \frac{\partial \phi}{\partial x_i} &= \frac{\partial}{\partial x_i} \left(\phi \frac{\partial \phi}{\partial x_i} \right) - \phi \left(\frac{\partial^2 \phi}{\partial x_i \partial x_i} \right) = \frac{\partial}{\partial x_i} \left(\phi \frac{\partial \phi}{\partial x_i} \right) - 4\pi G \rho \phi, \end{aligned} \quad (17)$$

and applying the divergence theorem we obtain

$$\mathcal{V}_g(t) = \frac{1}{2} \int_{M_t} \rho(\mathbf{y}, t) \phi(\mathbf{y}, t) d^3\mathbf{y},$$

$$\mathcal{V}_g(t) = \frac{1}{2} \int_{M_t} \varrho(\mathbf{y}, t) \phi(\mathbf{y}, t) d^3\mathbf{y}, \quad (18)$$

where, as above, we have used the continuity of ϕ and its normal derivative across ∂M_t , and the surface integral at infinity again vanishes. To include this binding energy into Hamilton's principle, it will be necessary to transform eq.(18) to the following integral over the reference body

$$\mathcal{V}_g(t) = \frac{1}{2} \int_M \rho(\mathbf{x}) \zeta(\mathbf{x}, t) d\mathbf{x}.$$

$$\mathcal{V}_g(t) = \frac{1}{2} \int_M \rho(\mathbf{x}) \zeta(\mathbf{x}, t) d^3\mathbf{x}, \quad (19)$$

where we have used the definition of the referential gravitational potential in eq.(6) and the statement of conservation of mass during the deformation in eq.(8).

The equations of motion relative to inertial space

Having obtained an appropriate expression for the gravitational binding energy, the equations of motion can be readily obtained using Hamilton's principle. The Lagrangian for the problem is

$$\mathcal{L}(\mathbf{x}, t, \varphi, \mathbf{v}, \mathbf{F}) = \frac{1}{2} \rho(\mathbf{x}) \|\mathbf{v}\|^2 - W(\mathbf{x}, t, \mathbf{F}) - \frac{1}{2} \rho(\mathbf{x}) \zeta(\mathbf{x}, t),$$

$$\mathcal{L}(\mathbf{x}, t, \varphi, \mathbf{v}, \mathbf{F}) = \frac{1}{2} \rho(\mathbf{x}) \|\mathbf{v}\|^2 - W(\mathbf{x}, t, \mathbf{F}) - \frac{1}{2} \rho(\mathbf{x}) \zeta(\mathbf{x}, t), \quad (20)$$

where we note that the elastic strain energy is allowed to be time-dependent to accommodate a seismic source. An important point about this Lagrangian is that it is **non-local** due to the gravitational binding energy. This means that the value of the Lagrangian at a point and time cannot be expressed only in terms of the values of the various fields and their derivatives at this point. The physical reason for this non-locality is, of course, the **action-at-a-distance** property of Newtonian gravitation.

The action for the body takes the usual form

$$\mathcal{S}(\varphi) = \int_0^T \int_M \mathcal{L}[\mathbf{x}, t, \varphi(\mathbf{x}, t), \mathbf{v}(\mathbf{x}, t), \mathbf{F}(\mathbf{x}, t)] d\mathbf{x} dt,$$

$$\mathcal{S}(\varphi) = \int_0^T \int_M \mathcal{L}[\mathbf{x}, t, \varphi(\mathbf{x}, t), \mathbf{v}(\mathbf{x}, t), \mathbf{F}(\mathbf{x}, t)] d^3\mathbf{x} dt, \quad (21)$$

and Hamilton's principle requires that this quantity be stationary about the true motion for all perturbations with fixed end points. In obtaining the Euler-Lagrange equations, however, we have to account for the non-local nature of the Lagrangian. As

ever, the mathematical statement of Hamilton's principle is ~~that the~~ vanishing of the functional derivative of the action

$$\langle D\mathcal{S}(\varphi)|\delta\varphi\rangle = 0,$$

$$\langle D\mathcal{S}(\varphi)|\delta\varphi\rangle = 0, \quad (22)$$

for all variations, $\delta\varphi$, subject to the usual fixed-endpoint conditions. Here the ~~the left hand-left-hand~~ side is, by definition, just the linear term in the expansion of $\mathcal{S}(\varphi + \delta\varphi)$ about φ . Looking at the form of the action, it is clear that this linear term takes the form

$$\begin{aligned} \langle D\mathcal{S}(\varphi)|\delta\varphi\rangle &= \int_0^T \int_M \left(\frac{\delta\mathcal{L}}{\delta\varphi_i} \delta\varphi_i + \frac{\delta\mathcal{L}}{\delta v_i} \delta v_i + \frac{\delta\mathcal{L}}{\delta F_{ij}} \delta F_{ij} \right) d\mathbf{x} dt. \\ \langle D\mathcal{S}(\varphi)|\delta\varphi\rangle &= \int_0^T \int_M \left(\frac{\delta\mathcal{L}}{\delta\varphi_i} \delta\varphi_i + \frac{\delta\mathcal{L}}{\delta v_i} \delta v_i + \frac{\delta\mathcal{L}}{\delta F_{ij}} \delta F_{ij} \right) d^3\mathbf{x} dt, \end{aligned} \quad (23)$$

where we have introduced the convenient notation $\frac{\delta\mathcal{L}}{\delta\varphi_i}$, $\frac{\delta\mathcal{L}}{\delta v_i}$ and $\frac{\delta\mathcal{L}}{\delta F_{ij}}$ for the factors multiplying the various perturbations. In the case of a local Lagrangian these factors reduce to normal partial derivatives, and hence it is only the term $\frac{\delta\mathcal{L}}{\delta\varphi_i}$ that requires additional thought. Applying the usual integration by parts steps, we arrive at the ~~Euler-Lagrange equation~~

$$\frac{\delta\mathcal{L}}{\delta\varphi_i} - \frac{\partial}{\partial t} \left(\frac{\delta\mathcal{L}}{\delta v_i} \right) - \frac{\partial}{\partial x_j} \left(\frac{\delta\mathcal{L}}{\delta F_{ij}} \right) = 0,$$

Euler-Lagrange equation

$$\frac{\delta\mathcal{L}}{\delta\varphi_i} - \frac{\partial}{\partial t} \left(\frac{\delta\mathcal{L}}{\delta v_i} \right) - \frac{\partial}{\partial x_j} \left(\frac{\delta\mathcal{L}}{\delta F_{ij}} \right) = 0, \quad (24)$$

along with the natural boundary condition

$$\frac{\delta\mathcal{L}}{\delta F_{ij}} \hat{n}_j = 0$$

$$\frac{\delta\mathcal{L}}{\delta F_{ij}} \hat{n}_j = 0 \quad (25)$$

on ∂M .

As in the non-gravitating case we have

$$\frac{\delta\mathcal{L}}{\delta v_i} = \rho v_i, \quad \frac{\delta\mathcal{L}}{\delta F_{ij}} = -\frac{\partial W}{\partial F_{ij}}$$

$$\frac{\delta \mathcal{L}}{\delta v_i} = \rho v_i, \quad \frac{\delta \mathcal{L}}{\delta F_{ij}} = -\frac{\partial W}{\partial F_{ij}}, \quad (26)$$

while in the ~~final~~ second problem set you will show that

$$\frac{\delta \mathcal{L}}{\delta \varphi_i} = \rho \gamma_i$$

$$\frac{\delta \mathcal{L}}{\delta \varphi_i} = \rho \gamma_i, \quad (27)$$

with γ_i the referential gravitational acceleration as defined above. The equations of motion, therefore, take the expected form

$$\rho \frac{\partial v_i}{\partial t} - \frac{\partial T_{ij}}{\partial x_j} - \rho \gamma_i = 0,$$

$$\rho \frac{\partial v_i}{\partial t} - \frac{\partial T_{ij}}{\partial x_j} - \rho \gamma_i = 0, \quad (28)$$

with $T_{ij} = \frac{\partial W}{\partial F_{ij}}$ the first ~~Piola-Kirchhoff~~ Piola-Kirchhoff stress tensor. The form of these equations is not, of course, unexpected. To the balance of linear momentum obtained previously, we have simply added the gravitational force acting on the body due to its own mass.

Use of a co-rotating reference frame

We have obtained the equations of motion for a self-gravitating body with respect to an **inertial reference** frame defined by the “fixed stars”, and these results could be applied directly to the Earth. As, however, the Earth undergoes a nearly steady rotation about its polar axis, it is useful to reformulate the equations of motion with respect to a **non-inertial frame that rotates with the Earth**.

To do this we express the motion as

$$\varphi_i(\mathbf{x}, t) = \Phi_i(t) + R_{ij}(t) \bar{\varphi}_j(\mathbf{x}, t),$$

$$\varphi_i(\mathbf{x}, t) = \Phi_i(t) + R_{ij}(t) \bar{\varphi}_j(\mathbf{x}, t), \quad (29)$$

where $\mathbf{R}(t)$ is a rotation matrix. Here ~~$\Phi(t)$ denotes~~ $\Phi(t)$ denotes the translation of the reference frame relative to inertial space, and $\mathbf{R}(t)$ changes of orientation. The term $\bar{\varphi}(\mathbf{x}, t)$ describes the motion relative to the non-inertial frame, and will be called the **relative motion**.

Differentiating eq. ~~(29)~~ (29) with respect to time we see that the velocity relative to inertial space can be written

$$v_i = \frac{d\Phi_i}{dt} + R_{ij} \left(\frac{\partial \bar{\varphi}_j}{\partial t} + R_{kl} \frac{dR_{kl}}{dt} \bar{\varphi}_l \right)$$

$$\underline{v_i = \frac{d\Phi_i}{dt} + R_{ij} \left(\frac{\partial \bar{\varphi}_j}{\partial t} + R_{kj} \frac{dR_{kl}}{dt} \bar{\varphi}_l \right)}, \quad (30)$$

where we have used the identity $R_{kj}R_{kl} = \delta_{jl}$ for rotation matrices. In fact, if we differentiate this latter identity we find

$$\underline{\frac{dR_{kj}}{dt} R_{kl} + R_{kj} \frac{dR_{kl}}{dt} = 0.}$$

$$\underline{\frac{dR_{kj}}{dt} R_{kl} + R_{kj} \frac{dR_{kl}}{dt} = 0,} \quad (31)$$

and hence conclude that $R_{kj} \frac{dR_{kl}}{dt}$ is an **anti-symmetric matrix**. It follows that we can use the alternating tensor to write

$$\underline{R_{kj} \frac{dR_{kl}}{dt} = \epsilon_{jkl} \Omega_k}$$

$$\underline{R_{kj} \frac{dR_{kl}}{dt} = \epsilon_{jml} \Omega_m}, \quad (32)$$

for a uniquely determined vector $\underline{\Omega}$. Using this result, the velocity becomes

$$\underline{v_i = \frac{d\Phi_i}{dt} + R_{ij} (\bar{v}_j + \epsilon_{jkl} \Omega_k \bar{\varphi}_l).}$$

$$\underline{v_i = \frac{d\Phi_i}{dt} + R_{ij} (\bar{v}_j + \epsilon_{jkl} \Omega_k \bar{\varphi}_l).} \quad (33)$$

Here $\bar{v}$ is the velocity as seen from the non-inertial frame, while $\underline{\Omega}$ is the **angular velocity of the frame**. Differentiating again and applying the same ideas, the acceleration relative to inertial space can be written

$$\underline{\frac{\partial v_i}{\partial t} = \frac{d^2 \Phi_i}{dt^2} + R_{ij} \left(\frac{\partial \bar{v}_j}{\partial t} + 2\epsilon_{jkl} \Omega_k \bar{v}_l + \epsilon_{jkl} \frac{d\Omega_k}{dt} \bar{\varphi}_l + \epsilon_{jkl} \epsilon_{lmn} \Omega_k \Omega_m \bar{\varphi}_n \right)}$$

$$\underline{\frac{\partial v_i}{\partial t} = \frac{d^2 \Phi_i}{dt^2} + R_{ij} \left(\frac{\partial \bar{v}_j}{\partial t} + 2\epsilon_{jkl} \Omega_k \bar{v}_l + \epsilon_{jkl} \frac{d\Omega_k}{dt} \bar{\varphi}_l + \epsilon_{jkl} \epsilon_{lmn} \Omega_k \Omega_m \bar{\varphi}_n \right).} \quad (34)$$

On the **right hand** **right hand** side we can identify various terms. The first is the acceleration of the frame relative to inertial space. Within the brackets we then see, relative to the non-inertial frame, accelerations associated with the motion along with the expected **Coriolis**, **Euler**, and **centrifugal** contributions.

We have not yet specified *how* the non-inertial frame is to be chosen. Within seismology the most common approach is to take

$$\underline{\frac{d\Phi}{dt} = 0, \quad \frac{d\Omega}{dt} = 0,}$$

$$\frac{d\Phi}{dt} = 0, \quad \frac{d\Omega}{dt} = 0, \quad (35)$$

which corresponds to a frame whose **origin is static** relative to the “fixed stars” “fixed stars” but **rotates steadily**, with the angular velocity chosen so that the frame **approximately co-rotates with the Earth**. The advantage of this choice is its simplicity. But it does mean that orbital, rotational, and relative motion of the Earth are mixed up within the term $\bar{\varphi}$. A preferable approach is to use the so-called **Tisserand frame** which is defined such that the relative motion, $\bar{\varphi}$ carries no net linear **nor angular momentum**¹. **or angular momentum**¹ But we will stick to the simpler and more common choice in this course.

To complete the derivation of the equations of motion in the non-inertial frame, we note from eq.(29) that

$$F_{ij} = R_{ik} \bar{F}_{kj}$$

(29) that

$$F_{ij} = R_{ik} \bar{F}_{kj}, \quad (36)$$

with $\bar{F}$ the deformation gradient associated with relative motion. Due to material frame indifference, we then have $W(\mathbf{x}, \mathbf{F}) = W(\mathbf{x}, \bar{\mathbf{F}})$ as should be physically expected. Using the chain rule, we find

$$\begin{aligned} \bar{T}_{ij} &= \frac{\partial W}{\partial \bar{F}_{ij}}(\bar{\mathbf{F}}) = \frac{\partial}{\partial \bar{F}_{ij}} W(\mathbf{R}\bar{\mathbf{F}}) = \frac{\partial W}{\partial F_{kl}}(\mathbf{R}\bar{\mathbf{F}}) \frac{\partial}{\partial \bar{F}_{ij}} (R_{km} \bar{F}_{ml}) = R_{ki} T_{kj} \\ \bar{T}_{ij} &= \frac{\partial W}{\partial \bar{F}_{ij}}(\bar{\mathbf{F}}) = \frac{\partial}{\partial \bar{F}_{ij}} W(\mathbf{R}\bar{\mathbf{F}}) = \frac{\partial W}{\partial F_{kl}}(\mathbf{R}\bar{\mathbf{F}}) \frac{\partial}{\partial \bar{F}_{ij}} (R_{km} \bar{F}_{ml}) = R_{ki} T_{kj}, \end{aligned} \quad (37)$$

with the **left hand left-hand** side the stress associated with $\bar{\varphi}$. Passing the rotation matrix to the other side, we have then obtained the useful identity

$$T_{ij} = R_{ik} \bar{T}_{kj}$$

$$T_{ij} = R_{ik} \bar{T}_{kj}, \quad (38)$$

between the first **Piola-Kirchhoff Piola-Kirchhoff** stress relative to the inertial and non-inertial reference frames. Finally, we put eq.(29 (29)) into eq.(10 (10)) to find the relationship

$$\gamma_i = R_{ij} \bar{\gamma}_i$$

$$\gamma_i = R_{ij} \bar{\gamma}_j, \quad (39)$$

between the referential gravitational acceleration in the two frames, with $\bar{\gamma}$ defined exactly as in eq.(10 (10)) but with $\bar{\varphi}$ replacing φ . Using these various results we have

¹ See “On the elastodynamics of rotating planets” by Maitra & Al-Attar (2024) for details.

¹ See Maitra & Al-Attar (2024) for details.

shown that the equations of motion are equivalent to

$$\rho \left(\frac{\partial \bar{v}_i}{\partial t} + 2\epsilon_{ijk}\Omega_j \bar{v}_k + \epsilon_{ijk}\epsilon_{klm}\Omega_j \Omega_l \bar{\varphi}_m \right) - \frac{\partial \bar{T}_{ij}}{\partial x_j} - \rho \bar{\gamma}_i = 0.$$

$$\rho \left(\frac{\partial \bar{v}_i}{\partial t} + 2\epsilon_{ijk}\Omega_j \bar{v}_k + \epsilon_{ijk}\epsilon_{klm}\Omega_j \Omega_l \bar{\varphi}_m \right) - \frac{\partial \bar{T}_{ij}}{\partial x_j} - \rho \bar{\gamma}_i = 0. \quad (40)$$

These equations are identical in form to those in an inertial frame except for the inclusion of the Coriolis and centrifugal accelerations – the Euler acceleration has been removed because we have restricted attention to a steadily rotating frame. From this point on we will consistently work in the approximately co-rotating frame, and hence will simplify notations by dropping overlines on terms linked to the relative motion.

Linearised equations of motion

As ever, our main concern is with the linearised equations of motion about an equilibrium state. Strictly, we should use the term **relative equilibrium**, as the body is now undergoing a steadily-steady rotation relative to inertial space. We consider a disturbance to the body caused by a stress glut, and hence take the strain energy to be

$$W(\mathbf{x}, t, \mathbf{F}) = U(\mathbf{x}, \mathbf{C}) - s \frac{1}{2} \bar{S}_{ij}(\mathbf{x}, t) C_{ij}$$

$$W(\mathbf{x}, t, \mathbf{F}) = U(\mathbf{x}, \mathbf{C}) - s \frac{1}{2} \bar{S}_{ij}(\mathbf{x}, t) C_{ij}, \quad (41)$$

with s a perturbation parameter, and $\bar{S}$ a given stress glut. As in Lecture 15, the elastic tensor that appears below is the second derivative of the time-independent part $U(\mathbf{x}, \mathbf{C})$ of this strain energy function. In a now standard manner we can obtain linearised equations of motion for the displacement vector defined as the first-order term in the expansion

$$\varphi_i(\mathbf{x}, t) = x_i + s u_i(\mathbf{x}, t) + O(s^2),$$

$$\varphi_i(\mathbf{x}, t) = x_i + s u_i(\mathbf{x}, t) + O(s^2). \quad (42)$$

At zeroth-order zeroth order, the equations of motion take the form

$$\rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m - \frac{\partial T_{ij}^0}{\partial x_j} - \rho \gamma_i^0 = 0,$$

$$\rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m - \frac{\partial T_{ij}^0}{\partial x_j} - \rho \gamma_i^0 = 0, \quad (43)$$

where T_{ij}^0 and γ_i^0 denote the equilibrium values of the stress tensor and gravitational acceleration. We will have more to say about these equations in the following lecture,

but for the moment note that there must be a balance between the centrifugal ~~and~~ and gravitational forces on the body and ~~a~~ the resulting equilibrium stress; the simplifying assumption of a stress-free equilibrium is no longer an option.

From the first-order term in the expansion we arrive at the linearised equations of motion for the displacement field

$$\rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk}\Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l u_m \right) - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) - \rho\gamma_i^1 = -\frac{\partial \bar{S}_{ij}}{\partial x_j},$$

$$\rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk}\Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l u_m \right) - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) - \rho\gamma_i^1 = -\frac{\partial \bar{S}_{ij}}{\partial x_j}, \quad (44)$$

where γ_i^1 denotes the first-order perturbation to the referential gravitational ~~potential~~ acceleration. Using a binomial expansion, this term can be written explicitly as

$$\gamma_i^1 = -G \int_M \rho' \left(\frac{\delta_{ij}}{[(x_k - x'_k)(x_k - x'_k)]^{3/2}} - \frac{3(x_i - x'_i)(x_j - x'_j)}{[(x_k - x'_k)(x_k - x'_k)]^{5/2}} \right) (u_j - u'_j) dx',$$

$$\gamma_i^1 = -G \int_M \rho' \left(\frac{\delta_{ij}}{[(x_k - x'_k)(x_k - x'_k)]^{3/2}} - \frac{3(x_i - x'_i)(x_j - x'_j)}{[(x_k - x'_k)(x_k - x'_k)]^{5/2}} \right) (u_j - u'_j) d^3x', \quad (45)$$

where it is understood that primed terms are functions of the integration variable. The boundary conditions for the problem take the expected form

$$A_{ijkl} \frac{\partial u_k}{\partial x_l} \hat{n}_j = \bar{S}_{ij} \hat{n}_j.$$

$$A_{ijkl} \frac{\partial u_k}{\partial x_l} \hat{n}_j = \bar{S}_{ij} \hat{n}_j. \quad (46)$$

The key point is that the inclusion of gravitation and rotation into the problem only requires some additional terms in the equations of motion considered so far. It is notable, however, that these are no longer partial differential equations, but **integro partial differential integro-differential equations**, with both derivatives and integrals of the unknown functions occurring within the equations of motion. Again, this feature derives from the ~~action-at-a-distance~~ action-at-a-distance property of Newtonian gravitation.

The neglect of self-gravitation and rotation

We have obtained the linearised equations of motion within a steadily rotating and self-gravitating body. It is natural to ask if and why it has been reasonable to neglect these physical phenomena up till this point. To do so, we will perform a scaling analysis of the equations.

We assume that the displacement vector field is characterised by a typical amplitude $\|\mathbf{u}\| \sim U$, a length-scale $\|\mathbf{u}\| \sim U$, a length scale L and time-scale a time scale

T. The following ~~order of magnitude~~ order-of-magnitude scaling estimates are then immediate

$$\begin{aligned} \underline{\rho \frac{\partial^2 u_i}{\partial t^2}} &\sim \underline{\frac{\bar{\rho} U}{T^2}}, \\ \underline{\rho \epsilon_{ijk} \Omega_j \frac{\partial u_k}{\partial t}} &\sim \underline{\frac{\bar{\rho} \Omega U}{T}}, \\ \underline{\rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l u_m} &\sim \underline{2 \bar{\rho} \Omega^2 U}, \\ \underline{\frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right)} &\sim \underline{\frac{\bar{\mu} U}{L^2}}, \end{aligned}$$

$$\underline{\rho \frac{\partial^2 u_i}{\partial t^2}} \sim \underline{\frac{\bar{\rho} U}{T^2}}, \quad (47)$$

$$\underline{\rho \epsilon_{ijk} \Omega_j \frac{\partial u_k}{\partial t}} \sim \underline{\frac{\bar{\rho} \Omega U}{T}}, \quad (48)$$

$$\underline{\rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l u_m} \sim \underline{2 \bar{\rho} \Omega^2 U}, \quad (49)$$

$$\underline{\frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right)} \sim \underline{\frac{\bar{\mu} U}{L^2}}, \quad (50)$$

where $\bar{\rho}$ and $\bar{\mu}$ denote typical values of density and elastic ~~modulii~~ moduli within the body, while Ω is the magnitude of the rotation vector. The scaling estimate for the gravitational term within the equations of motion requires more thought. The referential gravitational potential at equilibrium is given by

$$\begin{aligned} \underline{\zeta_0} &= -G \int_M \frac{\rho'}{[(x_k - x'_k)(x_k - x'_k)]^{1/2}} d\mathbf{x}', \\ \underline{\zeta_0} &= -G \int_M \frac{\rho'}{[(x_k - x'_k)(x_k - x'_k)]^{1/2}} d^3 \mathbf{x}'. \end{aligned} \quad (51)$$

Taking the gradient of this expression twice, we obtain

$$\begin{aligned} \underline{\frac{\partial \zeta_0}{\partial x_i \partial x_j}} &= G \int_M \rho' \left(\frac{\delta_{ij}}{[(x_k - x'_k)(x_k - x'_k)]^{3/2}} - \frac{3(x_i - x'_i)(x_j - x'_j)}{[(x_k - x'_k)(x_k - x'_k)]^{5/2}} \right) d\mathbf{x}', \\ \underline{\frac{\partial^2 \zeta_0}{\partial x_i \partial x_j}} &= G \int_M \rho' \left(\frac{\delta_{ij}}{[(x_k - x'_k)(x_k - x'_k)]^{3/2}} - \frac{3(x_i - x'_i)(x_j - x'_j)}{[(x_k - x'_k)(x_k - x'_k)]^{5/2}} \right) d^3 \mathbf{x}', \end{aligned} \quad (52)$$

and using ~~the Poisson~~ Poisson's equation $\nabla^2 \zeta_0 = 4\pi G \rho$ along with eq. ~~(45)~~ (45) we arrive at the desired scaling estimate

$$\underline{\rho \gamma_i^1} \sim \underline{4\pi G \bar{\rho}^2 U}$$

$$\rho \gamma_i^1 \sim 4\pi G \bar{\rho}^2 U. \quad (53)$$

Using these results, we can quantify the relative importance of the terms within the force balance. First we have

$$\frac{\text{gravitational forces}}{\text{elastic forces}} = \frac{4\pi G \bar{\rho}^2 U}{\frac{\bar{\mu} U}{L^2}} = \frac{4\pi G \bar{\rho} L^2}{\bar{v}^2},$$

$$\frac{\text{gravitational forces}}{\text{elastic forces}} = \frac{4\pi G \bar{\rho}^2 U}{\frac{\bar{\mu} U}{L^2}} = \frac{4\pi G \bar{\rho} L^2}{\bar{v}^2}, \quad (54)$$

where $\bar{v} = \sqrt{\frac{\bar{\mu}}{\bar{\rho}}} \bar{v} = \sqrt{\bar{\mu}/\bar{\rho}}$ represents a typical elastic wave speed in the model. This ratio scales like L^2 where L is the ~~characteristics length-scale~~ characteristic length scale of the deformation. We conclude that for deformations occurring over sufficiently small ~~length-scales~~ length scales, gravitational forces will be negligible relative to elastic ones. We should then ask whether gravitational forces will ever be of importance within the Earth? To answer this, we note that the largest-scale deformation within the Earth can have L equal to its average radius ~~6371-km~~ 6371 km. Recalling that ~~$G = 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$~~ $G = 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$, and taking as typical values ~~$\bar{\rho} = 5000 \text{ kg m}^{-3}$ and $\bar{v} = 8000 \text{ ms}^{-1}$~~ we find

$$\frac{\text{gravitational forces}}{\text{elastic forces}} \sim 3.$$

$\bar{\rho} = 5000 \text{ kg m}^{-3}$ and $\bar{v} = 8000 \text{ m s}^{-1}$, we find

$$\frac{\text{gravitational forces}}{\text{elastic forces}} \sim 3. \quad (55)$$

For such planetary scale deformations, it follows that gravitational forces are of the same order of magnitude as elastic forces, and so cannot be neglected within the dynamics.

Next, we consider the ratio of the Coriolis and inertial terms, and find

$$\frac{\text{Coriolis term}}{\text{inertial term}} = \frac{\frac{\bar{\rho} \Omega U}{T}}{\frac{\bar{\rho} U}{T^2}} = \Omega T$$

$$\frac{\text{Coriolis term}}{\text{inertial term}} = \frac{\frac{\bar{\rho} \Omega U}{T}}{\frac{\bar{\rho} U}{T^2}} = \Omega T. \quad (56)$$

It follows that the Coriolis terms will be negligible for motions whose ~~time-scale~~ time scale is short relative to the rotation period of the Earth. A similar conclusion is readily shown to hold for the centrifugal term. It is known that seismic deformation is limited to periods shorter than about one hour, and so rotational effects play a relatively small (but not negligible) role in the problem. There are, however, processes at longer periods, e.g. those associated with solid-Earth tides, for which rotation becomes very important.

What you need to know and be able to do

- (i) Understand the relevant parts of Newtonian gravity, including the manner in which the gravitational potential and acceleration can be transformed into quantities defined on the reference body.
- (ii) Know physically what the gravitational binding energy of a planet represents, and how it can be determined.
- (iii) Know how the gravitational binding energy is incorporated into the Lagrangian, and how the equations of motion can be obtained. The full derivation is too long to be examined, but you may be asked to reproduce parts of it given appropriate information and guidance.
- (iv) Perform scaling arguments to determine the relative size of terms ~~with in~~ within an equation, and to determine whether some of them might be negligible. This is a technique required throughout this course, and you should ~~insure~~ ensure you are happy with it.

Lecture 22: Free Oscillations

David Al-Attar

Michaelmas Term, 2026

Outline and ~~Motivation~~motivation

Having obtained the equations of motion for a self-gravitating earth model, we begin our study of the Earth's **free oscillations** or **normal modes** (the terms are synonymous). The starting point is developing a concise operator notation that closely resembles what you will know from quantum mechanics. We analyse the associated **eigenvalue problem** in a non-rotating earth model. In particular, the forced response of such an earth model is obtained as an **eigenfunction expansion**, with the frequency-domain response having **resonances** at each of the **eigenfrequencies**. It is due to this fact that ~~the~~ observations of the eigenfrequencies can be made from sufficiently long ~~time-series~~time series, with these data providing important constraints on Earth structure. Indeed, such frequencies form the principal constraint on the Earth's density.

Linearised equations of motion

Within Lecture~~20~~ 21 we obtained the following linearised equations of motion

$$\rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk}\Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l u_m \right) - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) - \rho\gamma_i^1 = -\frac{\partial \bar{S}_{ij}}{\partial x_j},$$

$$\rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk}\Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l u_m \right) - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) - \rho\gamma_i^1 = -\frac{\partial \bar{S}_{ij}}{\partial x_j}, \quad (1)$$

for a self-gravitating and rotating elastic earth model relative to a co-rotating reference frame. Here ρ is the equilibrium density, u_i the displacement vector, Ω_i ~~is~~ the angular velocity, A_{ijkl} the elastic tensor, and γ_i^1 the perturbation to the gravitational acceleration_x which is given concretely by

$$\gamma_i^1 = -G \int_M \rho' \left(\frac{\delta_{ij}}{[(x_k - x'_k)(x_k - x'_k)]^{3/2}} - \frac{3(x_i - x'_i)(x_j - x'_j)}{[(x_k - x'_k)(x_k - x'_k)]^{5/2}} \right) (u_j - u'_j) d^3\mathbf{x}',$$

$$\gamma_i^1 = -G \int_M \rho' \left(\frac{\delta_{ij}}{[(x_k - x'_k)(x_k - x'_k)]^{3/2}} - \frac{3(x_i - x'_i)(x_j - x'_j)}{[(x_k - x'_k)(x_k - x'_k)]^{5/2}} \right) (u_j - u'_j) d^3\mathbf{x}', \quad (2)$$

where it is understood that primed terms are functions of the integration variable, and finally $\bar{S}_{ij}$ is the stress glut associated with a seismic source. The boundary conditions for the problem are

$$\begin{aligned} A_{ijkl} \frac{\partial u_k}{\partial x_l} \hat{n}_j &= \bar{S}_{ij} \hat{n}_j \\ \underline{A_{ijkl} \frac{\partial u_k}{\partial x_l} \hat{n}_j} &= \underline{\bar{S}_{ij} \hat{n}_j} \end{aligned} \quad (3)$$

on ∂M . While these equations express quite simple physics, they do have a complicated form, and hence it will help things greatly to find a simpler notation.

As a first step to doing this, we will express the equations in what is known as the **weak form**. Letting w_i be an arbitrary function we find from eq.(1) that

$$\int_M \left[\rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk} \Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l u_m \right) - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} - \bar{S}_{ij} \right) - \rho \gamma_i^1 \right] w_i^* d^3 \mathbf{x} = 0.$$

(1) that

$$\int_M \left[\rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk} \Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l u_m \right) - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} - \bar{S}_{ij} \right) - \rho \gamma_i^1 \right] w_i^* d^3 \mathbf{x} = 0, \quad (4)$$

where we note that $\mathbf{w}$ has been allowed to be complex-valued and $*$ denotes a complex conjugate^{1,1}. Performing an integration by parts and using the boundary conditions this becomes

$$\int_M \left[\rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk} \Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l u_m - \gamma_i^1 \right) w_i^* + \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} - \bar{S}_{ij} \right) \frac{\partial w_i^*}{\partial x_j} \right] d^3 \mathbf{x} = 0.$$

$$\int_M \left[\rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk} \Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l u_m - \gamma_i^1 \right) w_i^* + \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} - \bar{S}_{ij} \right) \frac{\partial w_i^*}{\partial x_j} \right] d^3 \mathbf{x} = 0. \quad (5)$$

We have obtained this result for arbitrary $\mathbf{w}$ whenever $\mathbf{u}$ satisfies eq.(1) and (33). Conversely, the argument can be reversed to show that if eq.(5) holds for all $\mathbf{w}$, then the field $\mathbf{u}$ must satisfy eq.(1) and (33), this being the weak form of the problem.

Eq.(5) involves several terms that depend linearly on $\mathbf{u}$ and conjugate-linearly on $\mathbf{w}$, with such objects known as **sesquilinear forms**^{2,2}. For example, if we define

$$\langle \mathbf{w} | P | \mathbf{u} \rangle = \int_M \rho w_i^* u_i d^3 \mathbf{x},$$

¹ The reason for this generality is that later we will consider an associated eigenvalue problem, and in this context complex-valued fields naturally occur.

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² The term "sesqui" is used instead of "bilinear" to indicate that the form is linear in one argument and conjugate linear in the other.

² The prefix "sesqui" (one and a half) is used instead of "bi" to indicate that the form is linear in one argument and conjugate-linear in the other.

$$\langle \mathbf{w} | P | \mathbf{u} \rangle = \int_M \rho w_i^* u_i d^3 \mathbf{x}, \quad (6)$$

then we can write the acceleration term in the weak form as

$$\begin{aligned} & \langle \mathbf{w} | P | \dot{\mathbf{u}} \rangle. \\ & \langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle, \end{aligned} \quad (7)$$

where ~~time-derivatives~~ time derivatives are denoted using dots. Hopefully the notation here should remind you of bra-ket notation. We will call $\langle \mathbf{w} | P | \mathbf{u} \rangle$ the **kinetic energy form**, and note that it may be associated with an operator, P , that acts on displacement vectors by $(P\mathbf{u})_i = \rho u_i$. In a similar manner we define

$$\begin{aligned} & \langle \mathbf{w} | W | \mathbf{u} \rangle = \int_M 2\rho \epsilon_{ijk} w_i^* \Omega_j u_k d^3 \mathbf{x} \\ & \langle \mathbf{w} | W | \mathbf{u} \rangle = \int_M 2\rho \epsilon_{ijk} w_i^* \Omega_j u_k d^3 \mathbf{x}, \end{aligned} \quad (8)$$

which we call the **Coriolis form**, this being associated with an operator, W , defined by $(W\mathbf{u})_i = 2\rho \epsilon_{ijk} \Omega_j u_k$. We finally have the **potential energy form**

$$\begin{aligned} & \langle \mathbf{w} | H | \mathbf{u} \rangle = \int_M \left(\rho \epsilon_{ijk} \epsilon_{klm} w_i^* \Omega_j \Omega_l u_m + A_{ijkl} \frac{\partial w_i^*}{\partial x_j} \frac{\partial u_k}{\partial x_l} - \rho w_i^* \gamma_i^1 \right) d^3 \mathbf{x}, \\ & \langle \mathbf{w} | H | \mathbf{u} \rangle = \int_M \left(\rho \epsilon_{ijk} \epsilon_{klm} w_i^* \Omega_j \Omega_l u_m + A_{ijkl} \frac{\partial w_i^*}{\partial x_j} \frac{\partial u_k}{\partial x_l} - \rho w_i^* \gamma_i^1 \right) d^3 \mathbf{x}, \end{aligned} \quad (9)$$

and write the force term as

$$\begin{aligned} & \langle \mathbf{w} | \mathbf{f} \rangle = \int_M \bar{S}_{ij} \frac{\partial w_i^*}{\partial x_j} d^3 \mathbf{x}. \\ & \langle \mathbf{w} | \mathbf{f} \rangle = \int_M \bar{S}_{ij} \frac{\partial w_i^*}{\partial x_j} d^3 \mathbf{x}. \end{aligned} \quad (10)$$

In this manner, we can concisely express the weak form of the equations of motion as

$$\begin{aligned} & \langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle + \langle \mathbf{w} | W | \dot{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \mathbf{u} \rangle = \langle \mathbf{w} | \mathbf{f} \rangle, \\ & \langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle + \langle \mathbf{w} | W | \dot{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \mathbf{u} \rangle = \langle \mathbf{w} | \mathbf{f} \rangle, \end{aligned} \quad (11)$$

which is to hold for all **test functions**, $\mathbf{w}$. From this equation it would be very tempting to write the problem in operator form as

$$\begin{aligned} & P\ddot{\mathbf{u}} + W\dot{\mathbf{u}} + H\mathbf{u} = \mathbf{f}. \\ & P\ddot{\mathbf{u}} + W\dot{\mathbf{u}} + H\mathbf{u} = \mathbf{f}. \end{aligned} \quad (12)$$

Here it is, however, important to remember that the problem's boundary conditions are incorporated within the potential energy form and force term. As a result, we should see eq. (12) as only a formal notation; the weak form incorporates the boundary conditions directly, and is generally to be preferred.

Properties of the sesquilinear forms

If we look at the kinetic energy form, then it is clear that

$$\begin{aligned} \langle \mathbf{w} | P | \mathbf{u} \rangle &= \langle \mathbf{u} | P | \mathbf{w} \rangle^*, \\ \langle \mathbf{w} | P | \mathbf{u} \rangle &= \langle \mathbf{u} | P | \mathbf{w} \rangle^*, \end{aligned} \quad (13)$$

and we say that the form is **self-adjoint**. Equivalently we can say that the associated operator, P , is self-adjoint. We also note that $\langle \mathbf{u} | P | \mathbf{u} \rangle$ is positive for any non-zero vector field $\mathbf{u}$, and hence the kinetic energy form defines an **inner product**. In contrast, for the Coriolis form, we find that

$$\begin{aligned} \langle \mathbf{w} | W | \mathbf{u} \rangle &= \int_M 2\rho \epsilon_{ijk} w_i^* \Omega_j u_k \, d^3\mathbf{x} = - \left(\int_M 2\rho \epsilon_{ijk} u_i^* \Omega_j w_k \, d^3\mathbf{x} \right)^* = - \langle \mathbf{u} | W | \mathbf{w} \rangle^*, \\ \langle \mathbf{w} | W | \mathbf{u} \rangle &= \int_M 2\rho \epsilon_{ijk} w_i^* \Omega_j u_k \, d^3\mathbf{x} = - \left(\int_M 2\rho \epsilon_{ijk} u_i^* \Omega_j w_k \, d^3\mathbf{x} \right)^* = - \langle \mathbf{u} | W | \mathbf{w} \rangle^*, \end{aligned} \quad (14)$$

this meaning that it is **anti-self-adjoint**. This implies, in particular, that $\langle \mathbf{u} | W | \mathbf{u} \rangle$ is purely imaginary for any $\mathbf{u}$.

For the potential energy form it is readily checked that the term associated with elasticity is self-adjoint due to the hyperelastic symmetry $A_{ijkl} = A_{klij}$. Similarly for the centrifugal term we have

$$\begin{aligned} \int_M \rho \epsilon_{ijk} \epsilon_{klm} w_i^* \Omega_j \Omega_l u_m \, d^3\mathbf{x} &= \left(\int_M \rho \epsilon_{ijk} \epsilon_{klm} w_i \Omega_j \Omega_l u_m^* \, d^3\mathbf{x} \right)^* = \left(\int_M \rho \epsilon_{ijk} \epsilon_{klm} u_i^* \Omega_j \Omega_l w_m \, d^3\mathbf{x} \right)^*, \\ \int_M \rho \epsilon_{ijk} \epsilon_{klm} w_i^* \Omega_j \Omega_l u_m \, d^3\mathbf{x} &= \left(\int_M \rho \epsilon_{ijk} \epsilon_{klm} w_i \Omega_j \Omega_l u_m^* \, d^3\mathbf{x} \right)^* = \left(\int_M \rho \epsilon_{ijk} \epsilon_{klm} u_i^* \Omega_j \Omega_l w_m \, d^3\mathbf{x} \right)^*, \end{aligned} \quad (15)$$

where to obtain the final equality we have just relabelled and permuted the indices appropriately. What takes more work is to show that

$$\begin{aligned} \int \rho w_i^* \gamma_i^1(\mathbf{u}) \, d^3\mathbf{x} &= \int \rho \gamma_i^1(\mathbf{w})^* u_i \, d^3\mathbf{x}, \\ \int_M \rho w_i^* \gamma_i^1(\mathbf{u}) \, d^3\mathbf{x} &= \int_M \rho \gamma_i^1(\mathbf{w})^* u_i \, d^3\mathbf{x}, \end{aligned} \quad (16)$$

where we have included arguments to show which displacement vector the gravitational acceleration term corresponds to. **Proof** The proof of this identity **not difficult is not difficult**, but nor is it enlightening. The end result is that the potential energy form is also self-adjoint.

Eigenfunctions and eigenvalues in a non-rotating earth model

Recall from the scaling arguments in Lecture 20–21 that while self-gravitation is important for ~~low-frequency~~ low-frequency seismology, rotation remains only a relatively small effect. For the moment let us, therefore, suppose that the earth model is non-rotating. The equations of motion then read in weak form

$$\begin{aligned} \langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \mathbf{u} \rangle &= \langle \mathbf{w} | \mathbf{f} \rangle, \\ \langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \mathbf{u} \rangle &= \langle \mathbf{w} | \mathbf{f} \rangle, \end{aligned} \quad (17)$$

which is to hold for any $\mathbf{w}$, and where it is understood that the centrifugal contribution to the potential energy form has been removed.

Suppose first that we set the stress glut equal to zero, and consider a time-harmonic displacement field

$$\begin{aligned} u_i(\mathbf{x}, t) &= s_i(\mathbf{x}) e^{i\omega t} \\ u_i(\mathbf{x}, t) &= s_i(\mathbf{x}) e^{i\omega t}, \end{aligned} \quad (18)$$

with $\mathbf{s}$ the spatial part of the solution, and ω an angular frequency. Putting this into eq. (17) we arrive at

$$\begin{aligned} -\omega^2 \langle \mathbf{w} | P | \mathbf{s} \rangle + \langle \mathbf{w} | H | \mathbf{s} \rangle &= 0. \\ -\omega^2 \langle \mathbf{w} | P | \mathbf{s} \rangle + \langle \mathbf{w} | H | \mathbf{s} \rangle &= 0. \end{aligned} \quad (19)$$

This equation has the trivial solution, $\mathbf{s} = 0$, but it is interesting to ask whether it has any non-trivial solutions. As you no doubt recognise, this is an eigenvalue problem with $\mathbf{s}$ the eigenfunction and ω the eigenfrequency. Note, in particular, that the boundary conditions for the problem are built into the weak form of the equations of motion, and hence these eigenfunctions satisfy traction-free boundary conditions on ∂M .

Due to the self-adjointness of the kinetic and potential energy forms, we can quickly establish some expected properties of the eigenfunctions and eigenfrequencies. Let $\mathbf{s}_1$ be an eigenfunction associated with eigenfrequency ω_1 , and $\mathbf{s}_2$ an eigenfunction associated with eigenfrequency ω_2 . By definition we then have

$$\begin{aligned} -\omega_1^2 \langle \mathbf{w} | P | \mathbf{s}_1 \rangle + \langle \mathbf{w} | H | \mathbf{s}_1 \rangle &= 0, \\ -\omega_1^2 \langle \mathbf{w} | P | \mathbf{s}_1 \rangle + \langle \mathbf{w} | H | \mathbf{s}_1 \rangle &= 0, \end{aligned} \quad (20)$$

for any $\mathbf{w}$. Taking this test function equal to $\mathbf{s}_2$, and using the self-adjointness of the forms, it follows that

$$\begin{aligned} 0 &= -\omega_1^2 \langle \mathbf{s}_2 | P | \mathbf{s}_1 \rangle + \langle \mathbf{s}_2 | H | \mathbf{s}_1 \rangle = -\omega_1^2 \langle \mathbf{s}_1 | P | \mathbf{s}_2 \rangle^* + \langle \mathbf{s}_1 | H | \mathbf{s}_2 \rangle^*. \\ 0 &= -\omega_2^2 \langle \mathbf{s}_2 | P | \mathbf{s}_1 \rangle + \langle \mathbf{s}_2 | H | \mathbf{s}_1 \rangle = -\omega_2^2 \langle \mathbf{s}_1 | P | \mathbf{s}_2 \rangle^* + \langle \mathbf{s}_1 | H | \mathbf{s}_2 \rangle^*. \end{aligned} \quad (21)$$

Using the eigenvalue equation for $\mathbf{s}_2$ we know that

$$\langle \mathbf{s}_1 | H | \mathbf{s}_2 \rangle = \omega_2^2 \langle \mathbf{s}_1 | P | \mathbf{s}_2 \rangle,$$

$$\langle \mathbf{s}_1 | H | \mathbf{s}_2 \rangle = \omega_2^2 \langle \mathbf{s}_1 | P | \mathbf{s}_2 \rangle, \quad (22)$$

and hence from eq. (21) we have shown that

$$\underbrace{(\omega_1^2 - \omega_2^{2*}) \langle \mathbf{s}_1 | P | \mathbf{s}_2 \rangle^* = 0.}_{(\omega_1^2 - \omega_2^{2*}) \langle \mathbf{s}_1 | P | \mathbf{s}_2 \rangle^* = 0.} \quad (23)$$

Taking $\mathbf{s}_1 = \mathbf{s}_2$, and recalling that $\langle \mathbf{u} | P | \mathbf{u} \rangle$ is positive for any non-zero displacement field, we conclude that $\omega_1^{2*} = \omega_1^2$, which is to say that the **squared-eigenfrequencies** squared eigenfrequencies are real. Next, if we suppose that $\omega_1 \neq \omega_2$, $\omega_1^2 \neq \omega_2^2$, then we must have

$$\begin{aligned} \langle \mathbf{s}_1 | P | \mathbf{s}_2 \rangle &= 0 \\ \langle \mathbf{s}_1 | P | \mathbf{s}_2 \rangle &= 0, \end{aligned} \quad (24)$$

this meaning that eigenfunctions with different **squared-eigenfrequencies** squared eigenfrequencies are **orthogonal relative to the inner product defined by kinetic energy form**. Orthogonality relative to such a “**mass-weighted** mass-weighted” inner product should be familiar from your prior study of linear oscillations of finite-dimensional mechanical systems.

The following additional properties of these eigenfunctions and eigenfrequencies will be stated without proof^{3,3}

1. Each **squared-eigenfrequency** squared eigenfrequency is associated with a **finite-dimensional eigenspace**. Eigenfunctions within the same eigenspace need not be orthogonal, but we can always select a basis such that they are.
2. There is a **countable infinity** of **squared-eigenfrequencies** squared eigenfrequencies. Their values are **bounded from below**, and **accumulate only at infinity**. The latter point means that in a **finite-interval** finite interval there can only be a finite number of **squared-eigenfrequencies** squared eigenfrequencies.
3. The eigenfunctions form a **complete orthonormal basis** for vector fields in the earth model. This means that an arbitrary vector field can be expressed as a convergent sum of the eigenfunctions, with the necessary expansion coefficients readily determined due to their orthogonality.

To proceed, we let $k = 0, 1, \dots$ label the different squared eigenfrequencies, ω_k^2 . For each k , there is then a finite-dimensional eigenspace, and we use a second index m to label the elements of its basis. Note that the range of m can vary with k , but we leave this implicit within the notation. The m th eigenfunction belonging to the k th eigenspace will be written $|km\rangle$ in a manner familiar from quantum mechanics.

³ ~~The validity of these results in solid earth models can be established rigorously based on the theory of elliptic partial differential operators. Their proof, and indeed general validity, within models containing in-viscid fluid regions like the outer core remains an open problem.~~

³ The validity of these results in solid earth models can be established rigorously based on the theory of elliptic partial differential operators. Their proof, and indeed general validity, within models containing inviscid fluid regions like the outer core remains an open problem.

These eigenfunctions can be orthonormalised relative to the kinetic energy form, this meaning that

$$\begin{aligned} \langle km|P|k'm'\rangle &= \delta_{kk'}\delta_{mm'}, \\ \langle km|P|k'm'\rangle &= \delta_{kk'}\delta_{mm'}, \end{aligned} \quad (25)$$

for all possible indices. Using this notation, we can then express the completeness of the eigenfunctions by writing an arbitrary vector field $\mathbf{u}$ that satisfies the necessary boundary conditions in the form

$$\begin{aligned} \mathbf{u} &= \sum_{km} \langle km|P|\mathbf{u}\rangle |km\rangle, \\ \mathbf{u} &= \sum_{km} \langle km|P|\mathbf{u}\rangle |km\rangle, \end{aligned} \quad (26)$$

where $\langle km|P|\mathbf{u}\rangle$ are the expansion coefficients of the (k, m) th term.

Eigenfunction expansions

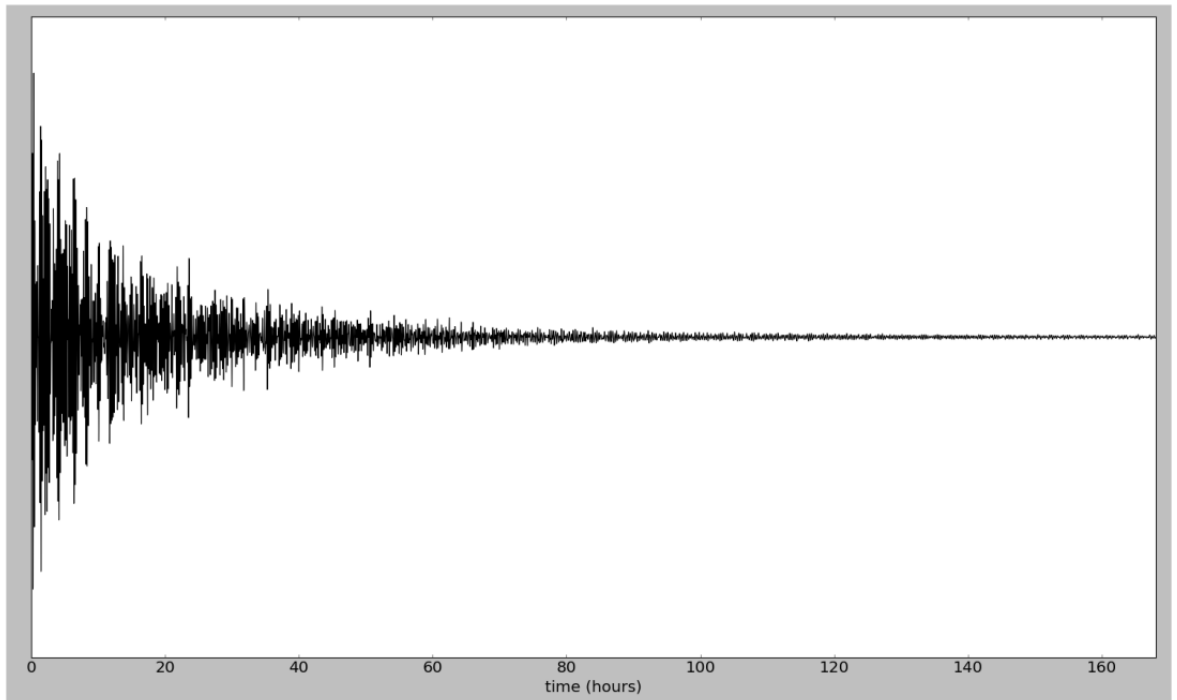


Fig. 1: A ~~seven-day-long-vertical-component~~ seven-day-long vertical-component seismogram following the ~~a~~ magnitude 8.2 earthquake in Bolivia in 1994. Note ~~also~~ that the overall amplitude decays in a roughly exponential fashion. This is due to attenuation within the Earth, but we ignore this effect within our current discussion.

Recall that the equations of motion in weak form are given by

$$\langle \mathbf{w}|P|\ddot{\mathbf{u}}\rangle + \langle \mathbf{w}|H|\mathbf{u}\rangle = \langle \mathbf{w}|\mathbf{f}\rangle,$$

$$\langle \mathbf{w} | P | \dot{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \mathbf{u} \rangle = \langle \mathbf{w} | \mathbf{f} \rangle, \quad (27)$$

for any ~~test-function~~ test function, $\mathbf{w}$. These equations can be usefully studied in the ~~frequency-domain~~ frequency domain, where we use the convention

$$\tilde{\mathbf{u}}(\mathbf{x}, \omega) = \int_0^\infty \mathbf{u}(\mathbf{x}, t) e^{-i\omega t} dt, \quad \mathbf{u}(\mathbf{x}, t) = \frac{1}{2\pi} \int_{-\infty}^\infty \tilde{\mathbf{u}}(\mathbf{x}, \omega) e^{i\omega t} d\omega, \quad (28)$$

noting that the initial conditions that $\mathbf{u}$ and $\dot{\mathbf{u}}$ vanish before time $t = 0$ ~~has~~ have been built into the forward transformation. It is readily seen that the Fourier transformed equations of motion take the form

$$-\omega^2 \langle \mathbf{w} | P | \tilde{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \tilde{\mathbf{u}} \rangle = \langle \mathbf{w} | \tilde{\mathbf{f}} \rangle. \quad (29)$$

Using the completeness of the eigenfunctions, we know that we can write

$$\tilde{\mathbf{u}} = \sum_{k'm'} \langle k'm' | P | \tilde{\mathbf{u}} \rangle | k'm' \rangle, \quad (30)$$

for some frequency-dependent coefficients, $\langle k'm' | P | \tilde{\mathbf{u}} \rangle$, that are to be determined. Putting this eigenfunction expansion into eq. ~~(29)~~ (29) we find

$$\sum_{k'm'} [-\omega^2 \langle \mathbf{w} | P | k'm' \rangle + \langle \mathbf{w} | H | k'm' \rangle] \langle k'm' | P | \tilde{\mathbf{u}} \rangle = \langle \mathbf{w} | \tilde{\mathbf{f}} \rangle. \quad (31)$$

which can be simplified using the eigenvalue problem satisfied by $|k'm'\rangle$ to read

$$\sum_{k'm'} (-\omega^2 + \omega_{k'}^2) \langle \mathbf{w} | P | k'm' \rangle \langle k'm' | P | \tilde{\mathbf{u}} \rangle = \langle \mathbf{w} | \tilde{\mathbf{f}} \rangle. \quad (32)$$

Finally, we take the test function equal to $|km\rangle$, and use the orthonormality condition in eq. ~~(25)~~ (25) to find that the (k, m) th coefficient of $\tilde{\mathbf{u}}$ is given by

$$\langle km | P | \tilde{\mathbf{u}} \rangle = \frac{\langle km | \tilde{\mathbf{f}} \rangle}{\omega_k^2 - \omega^2}$$

$$\langle km|P|\tilde{\mathbf{u}}\rangle = \frac{\langle km|\tilde{\mathbf{f}}\rangle}{\omega_k^2 - \omega^2}. \quad (33)$$

In this manner, we have obtained the **eigenfunction expansion solution**

$$\tilde{\mathbf{u}} = \sum_{km} \frac{\langle km|\tilde{\mathbf{f}}\rangle}{\omega_k^2 - \omega^2} |km\rangle, \quad (34)$$

$$\tilde{\mathbf{u}} = \sum_{km} \frac{\langle km|\tilde{\mathbf{f}}\rangle}{\omega_k^2 - \omega^2} |km\rangle,$$

for the frequency-domain problem. Inversion of this solution into the **time-domain** can be readily done using contour integration. Such techniques are not necessary for this course, but in the **final-second** problem set you will be given and make use of the resulting time-domain solution.

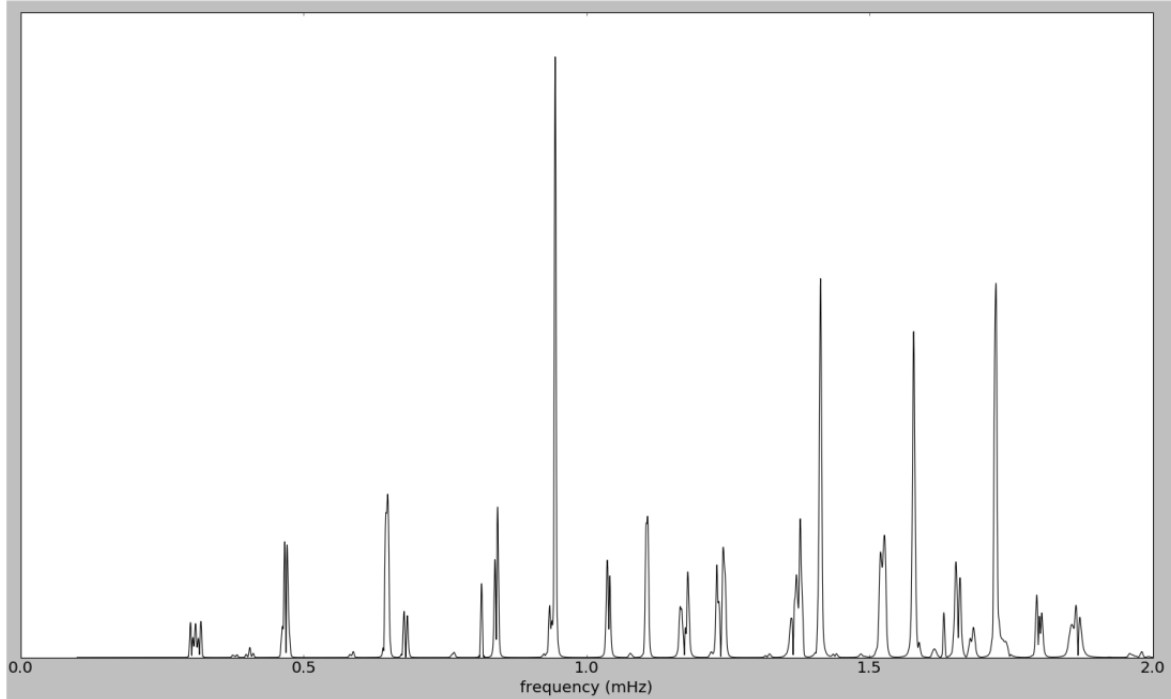


Fig. 2: The amplitude spectra corresponding to Fig. 1. In some cases individual mode peaks can be identified, but especially at **higher-frequencies** there is substantial overlap of the peaks due both to attenuation and an insufficiently long time series.

Looking at eq. (34), it is notable that there are singularities at $\pm\omega_k$ for each k . Supposing for the moment that the **squared-eigenfrequencies** are all positive, these singularities lie along the real line. Physically, we are seeing **resonances** in the frequency-domain response when the system is driven at one of its natural frequencies. From an actual seismic record we could take its Fourier transform, look for associated resonance peaks, and hence measure some of the eigenfrequencies of the Earth. These frequencies are determined solely by Earth structure, and hence

their measurement provides a possible means to avoid the **source-structure trade-off** that is ever present in seismology.

To consider such measurements further, it is important to note that recorded seismograms are not of an infinite length, and so their Fourier transform will not yield eq.(34 (34) exactly. In fact, within a more complete treatment there is also slight attenuation that shifts the singularities into the upper half-plane. To understand what happens with a finite-length time series, we can introduce a windowing function, $h(t)$, that is non-zero only in the range $[0, T]$, and so write the observed ~~time-series as~~

$$\underline{\mathbf{u}_{obs}(t) = h(t)\mathbf{u}(t)}$$

time series as

$$\underline{\mathbf{u}_{obs}(t) = h(t)\mathbf{u}(t)}, \quad (35)$$

where we neglect spatial arguments for simplicity. Taking the Fourier transform of such a ~~time-series~~ time series and applying the convolution theorem we find

$$\begin{aligned} \underline{\tilde{\mathbf{u}}_{obs}(\omega) \propto \int_{-\infty}^{\infty} \tilde{\mathbf{u}}(\omega') \tilde{h}(\omega' - \omega) d\omega',} \\ \underline{\underline{\tilde{\mathbf{u}}_{obs}(\omega) \propto \int_{-\infty}^{\infty} \tilde{\mathbf{u}}(\omega') \tilde{h}(\omega - \omega') d\omega',}} \end{aligned} \quad (36)$$

with the ~~right hand~~ right-hand side being proportional to the convolution of the two Fourier transforms. The end result is that the singularities of the Fourier-domain solution in eq.(34 (34) are smoothed out within real observations. The extent of the smoothing is controlled primarily by the length of the ~~time-serie~~ time series, with the support of $\tilde{h}$ being largely confined to an interval whose width scales like T^{-1} . This is, of course, a familiar result; longer ~~time-series~~ time series provide better resolution in the ~~frequency-domain~~ frequency domain.

It is for this reason that observational work on the Earth's free oscillations is concerned only with the largest earthquakes, the response to which can remain above background noise for many days. Even so, within observed spectra we cannot always identify individual mode peaks, and this is especially true in cases of near degeneracy that we will examine in more detail in the next lecture.

Gravitational stability

Within the previous section we assumed that the squared eigenfrequencies were all positive, with the result that each term in the mode sum is associated with two resonances on the real axis, and hence oscillatory contributions to the time-domain solution. If, instead, a squared eigenfrequency, ω_k^2 , ~~was~~ were negative then there would be two resonances on the imaginary axis, resulting in exponentially growing and decaying terms in the time-domain solution. If a squared eigenfrequency vanished, then there would be a ~~double-root~~ double root of the denominator at the origin, and this can be shown to ~~results~~ result in a term in the time-domain solution that grows linearly with time. The underlying issue here is whether the earth model is **stable to small perturbations**, with its behaviour being **encoded within the sign of its squared eigenfrequencies**.

Recalling the form of the eigenvalue problem

$$\begin{aligned} & \underline{-\omega_k^2 \langle \mathbf{w} | P | km \rangle + \langle \mathbf{w} | H | km \rangle = 0.} \\ & \underline{-\omega_k^2 \langle \mathbf{w} | P | km \rangle + \langle \mathbf{w} | H | km \rangle = 0,} \end{aligned} \quad (37)$$

we can set the test function equal to $|km\rangle$ to obtain

$$\begin{aligned} & \underline{\omega_k^2 = \langle km | H | km \rangle.} \\ & \underline{\omega_k^2 = \langle km | H | km \rangle,} \end{aligned} \quad (38)$$

where we have made use of the eigenfunction normalisation condition. It follows that the sign of the squared eigenfrequency is set by the potential energy form.

In the non-gravitating case, this term is given by

$$\begin{aligned} & \underline{\langle \mathbf{u} | H | \mathbf{u} \rangle = \int_M A_{ijkl} \frac{\partial u_i^*}{\partial x_j} \frac{\partial u_k}{\partial x_l} d^3 \mathbf{x},} \\ & \underline{\langle \mathbf{u} | H | \mathbf{u} \rangle = \int_M A_{ijkl} \frac{\partial u_i^*}{\partial x_j} \frac{\partial u_k}{\partial x_l} d^3 \mathbf{x},} \end{aligned} \quad (39)$$

which is real due to the hyperelastic symmetry. We know from the first problem set that in the absence of pre-stress the elastic tensor has the additional symmetries $A_{ijkl} = A_{jikl} = A_{ijlk}$, and hence the above expression becomes

$$\begin{aligned} & \underline{\langle \mathbf{u} | H | \mathbf{u} \rangle = \frac{1}{4} \int_M A_{ijkl} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^* \left(\frac{\partial u_k}{\partial x_l} + \frac{\partial u_l}{\partial x_k} \right) d^3 \mathbf{x},} \\ & \underline{\langle \mathbf{u} | H | \mathbf{u} \rangle = \frac{1}{4} \int_M A_{ijkl} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^* \left(\frac{\partial u_k}{\partial x_l} + \frac{\partial u_l}{\partial x_k} \right) d^3 \mathbf{x},} \end{aligned} \quad (40)$$

where we have explicitly symmetrised the spatial derivatives. We will say that the elastic tensor is pointwise stable at $\mathbf{x}$ if

$$\begin{aligned} & \underline{A_{ijkl}(\mathbf{x}) e_{ij}^* e_{kl} \geq k e_{ij}^* e_{ij}} \\ & \underline{A_{ijkl}(\mathbf{x}) e_{ij}^* e_{kl} \geq c_0 e_{ij}^* e_{ij},} \end{aligned} \quad (41)$$

for some constant $\underline{k} \gg \underline{0}$, $\underline{c_0} \geq 0$, and all symmetric tensors e_{ij} . If such a condition holds at all points in the earth model, then we say that the elastic tensor is **uniformly pointwise stable**. Assuming this is true, we then have

$$\underline{\langle \mathbf{u} | H | \mathbf{u} \rangle = \frac{1}{4} \int_M A_{ijkl} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^* \left(\frac{\partial u_k}{\partial x_l} + \frac{\partial u_l}{\partial x_k} \right) d^3 \mathbf{x} > 0,}$$

$$\langle \mathbf{u} | H | \mathbf{u} \rangle = \frac{1}{4} \int_M A_{ijkl} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^* \left(\frac{\partial u_k}{\partial x_l} + \frac{\partial u_l}{\partial x_k} \right) d^3 \mathbf{x} > 0, \quad (42)$$

so long as the linearised strain $e_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$ is not everywhere equal to zero. It is not difficult to show that e_{ij} vanishes identically for a non-trivial displacement vector if and only if it takes the form

$$\begin{aligned} u_i(\mathbf{x}) &= a_i + A_{ij}x_j, \\ u_i(\mathbf{x}) &= a_i + B_{ij}x_j, \end{aligned} \quad (43)$$

with a_i a constant vector, and A_{ij} an anti-symmetric matrix. Such a displacement vector corresponds to an **infinitesimal rigid body motion**, and hence this result makes physical sense if we recall the principle of material frame indifference. It can be shown that the pointwise stability condition holds in an isotropic medium if and only if the bulk and shear moduli are both positive, and this in turn implies that P- and S-waves have positive phase speeds , with the P-wave speed being faster with $\alpha > \beta$.

To summarise, we have seen that in the case of a non-gravitating earth model the squared-eigenfrequencies are all non-negative so long as the elastic tensor is uniformly pointwise stable. There is a six-dimensional eigenspace at zero-frequency associated with rigid body motions. Recalling, however, that the force term in the problem is

$$\begin{aligned} \langle \mathbf{w} | \mathbf{f} \rangle &= \int_M \bar{S}_{ij} \frac{\partial w_i^*}{\partial x_j} d^3 \mathbf{x}, \\ \langle \mathbf{w} | \mathbf{f} \rangle &= \int_M \bar{S}_{ij} \frac{\partial w_i^*}{\partial x_j} d^3 \mathbf{x}, \end{aligned} \quad (44)$$

with $\bar{S}_{ij}$ symmetric, we see that these **trivial modes** cannot be excited by a seismic source. Thus under a reasonable restriction on the elastic tensor, we see that a non-gravitating earth model is always stable to internal perturbations.

With gravitation added to the problem there remains a six-dimensional eigenspace at zero-frequency associated with rigid body motions. The issue of stability of the remaining modes is, however, complicated by the fact that the potential energy has a gravitational contribution. While it is possible to examine this problem quantitatively, it requires some fairly advanced techniques. The physically important point, however, is that **self-gravitation is an inherently destabilising force**. Consider, for example, a spherically symmetric and self-gravitating planet at equilibrium. If we could compress this planet uniformly, then there would be an increase in elastic potential energy. At the same time, the mass of the planet has been brought into a smaller volume, and hence the gravitational binding energy must decrease^{4, 4}. The net effect on the potential energy depends on the relative magnitude of these terms. If the elastic constants are sufficiently large, then the elastic term will dominate and the planet will be stable with respect to such a perturbation. But if gravity is sufficiently strong then the planet will keep pulling itself inwards until non-linear effects take over.

⁴ Recall that this is a negative quantity, and hence it has increased in magnitude.

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What you need to know and be able to do

- (i) The weak formulation of the equations of motion using ~~bra-ket-like~~ bra-ket-like notation, and the properties of the different sesquilinear forms.
- (ii) How to derive the reality of the ~~squared-eigenfrequencies~~ squared eigenfrequencies and orthogonality of eigenfunctions in distinct eigenspaces. State without proof the remaining important properties of the eigenvalue problem.
- (iii) How to obtain a solution of the forced equations of motion in a non-rotating planet as an eigenfunction expansion in the ~~frequency-domain~~ frequency domain. You do not need to know how to recover the time-domain solution, but may be asked to verify that such a solution works.
- (iv) That eigenfrequencies are associated with resonance peaks in the amplitude spectra obtained from seismograms, and why it is necessary to look at data from large earthquakes to obtain adequate resolution.
- (v) What it means for a non-rotating earth model to be stable in terms of the eigenfrequencies. You should understand the quantitative argument for the non-gravitating case, and the qualitative one when gravity is included.